

Agenda:

1.) Announcements:

- 9.2, 9.3, 9.4 HW due Wednesday
- Discussion Quiz 5 on Thursday over
- Exams will be returned on Thursday

2.) Section 11.1 (continued)

Convergence Let $\{a_n\}$ be a sequence

1.) We say that $\{a_n\}$ converges to L , written as
 $\lim_{n \rightarrow \infty} a_n = L$ if, for every $\varepsilon > 0$, there is a positive
integer N such that $|a_n - L| < \varepsilon$ for all $n \geq N$

2.) We say that $\{a_n\}$ diverges if $\lim_{n \rightarrow \infty} a_n$ does not exist
includes $-\infty$ or ∞

Theorem: If $f(x)$ is a real-valued function where
 $f(n) = a_n$ for all positive integers n and $\lim_{x \rightarrow \infty} f(x) = L$
then $\lim_{n \rightarrow \infty} f(n) = L$
rem numbers

$\uparrow$
Positive integers $\rightarrow \lim_{n \rightarrow \infty} a_n = L$

Activity: $\lim_{n \rightarrow \infty} \frac{3n^3 + 7n^2 + 1}{4n^3 + 8n + 8} \cdot \frac{\frac{1}{n^3}}{\frac{1}{n^3}}$
 $= \frac{\frac{3n^3}{n^3} + \frac{7n^2}{n^3} + \frac{1}{n^3}}{\frac{4n^3}{n^3} + \frac{8n^2}{n^3} + \frac{8}{n^3}} = \left(\frac{3}{4}\right)$

Example: $\lim_{n \rightarrow \infty} \cos(2\pi n)$

$\uparrow$
all integers ≥ 1

$$\lim_{n \rightarrow \infty} \cos(2\pi n) = \lim_{n \rightarrow \infty} 1 = 1$$

$\lim_{x \rightarrow \infty} \cos(2\pi x) \rightarrow \text{oscillates}$

✱ Can't conclude from limit of function the limit of the sequence.
If the limit of the function DNE.

Properties of Sequences

There are operations we can perform on two sequences

$$(1) \{a_n\} + \{b_n\} \Rightarrow \{a_1 + b_1, a_2 + b_2, \dots\} \\ = \{a_n + b_n\}$$

$$(2) c\{a_n\} \Rightarrow \{ca_n\}, \text{ where } c \text{ is a constant.}$$

$$(3) \{a_n\} \cdot \{b_n\} \Rightarrow \{a_n b_n\}$$

Activity: Let $\{a_n\} = \{(\frac{1}{3})^n\}$ and $\{b_n\} = \{(\frac{1}{4})^n\}$. Find:

$$(1.) \{a_n\} + \{b_n\} \Rightarrow \{(\frac{1}{3})^n + (\frac{1}{4})^n\} = \left\{ \frac{3^n + 4^n}{12^n} \right\}$$

$$(2.) 7 \cdot \{a_n\} \Rightarrow \left\{ \frac{7}{3^n} \right\} \text{ or } \left\{ 7 \left(\frac{1}{3} \right)^n \right\}$$

form: $a_1 r^{n-1}$
 $\Rightarrow (\frac{1}{3})^n$

$$(3.) \{a_n\} \cdot \{b_n\} \Rightarrow \left\{ \left(\frac{1}{12} \right)^n \right\} \text{ or } \left\{ \frac{1}{12^n} \right\}$$

$a_1 r^{n-1}$
 $1 \left(\frac{1}{12} \right)^n$

geometric sequences

Limit Laws

Suppose $\{a_n\}$ and $\{b_n\}$ converge. Let c be a constant.

$$1.) \lim_{n \rightarrow \infty} c a_n = c \lim_{n \rightarrow \infty} a_n$$

$$2.) \lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

$$3.) \lim_{n \rightarrow \infty} (a_n \cdot b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$$

$$4.) \lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}, \quad (\lim_{n \rightarrow \infty} b_n \neq 0)$$

Activity:

$$1.) \lim_{n \rightarrow \infty} \left(3 - \frac{n-1}{n+1} \right) = 2$$

$$2.) \lim_{n \rightarrow \infty} [(-1)^n + (-1)^{n+1}] = 0$$

$$3.) \lim_{n \rightarrow \infty} \frac{n^2}{e^n} = 0$$

$$\lim_{x \rightarrow \infty} \frac{x^2}{e^x} \stackrel{LH}{=} \frac{2x}{e^x} \stackrel{LH}{=} \frac{2}{e^x} = \frac{0}{\infty} = 0$$

Useful Theorems

Theorem Let $\{a_n\}$ be a geometric sequence where $a_n = a_1 r^{n-1}$ for constants a_1 and r

1.) if $-1 < r < 1$, then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_1 r^{n-1} = 0$

if $r > 1$, then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_1 r^{n-1} = \infty$

if $r = 1$, then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_1 r^{n-1} = a_1$

if $r = -1$, then $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_1 r^{n-1} \rightarrow DNE$

Activity

1.) $\lim_{n \rightarrow \infty} 5\left(\frac{2}{3}\right)^{n-1}$

2.) $\lim_{n \rightarrow \infty} 15$

3.) $\lim_{n \rightarrow \infty} (-2)^n$

Theorem: If $\lim_{n \rightarrow \infty} |a_n| = 0$, then $\lim_{n \rightarrow \infty} a_n = 0$

Example: $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n}$

~~7~~ ~~13~~ ~~28~~ ~~98~~ 9

Monotonic Sequence Theorem

Definitions: Let $\{a_n\}$ be a sequence

1.) It is called monotonically increasing if

2.) It is called monotonically decreasing if

3.) It is called bounded if

Example:

1.) $a_n = -2 + 3n$

2.) $b_n = \cos(n)$

3.) $a_n = 1 + \frac{1}{n} = \frac{n+1}{n}$

Monotonic Sequence Theorem: